Q1) [10 marks] Represent the following system as in Simulink

a. $y(t) = 4 + 0.6\,t^2 + 3\int y(t)dt$

b. $2Y''(t) = 6\,Y'(t)\,Z'(t) + 4Y(t)$ $\qquad Z''(t) = 3\,Z'(t)\,Y(t) + 0.8\,Z(t)$

Q2. [10 marks] Plot the output response for each of the following conditions



If r(t) = 1

a) when k = 1 ,T= 2 and a=0

b) when k = 1 ,T= 1 and a=0

c) when k = 1 ,T= 1 and a=1

d) when k =1 ,T=2 and a=1

e) when k =1 ,T=1 and a=5

Q1) [10 marks]

a) Determine the state space matrices A,B,C & D for a system given

$$Y'''(t) + 0.6Y''(t) + 2Y = 3r(t)$$

Parameters

A:

B:

C:

D:

b) Find the transfer function of above equation